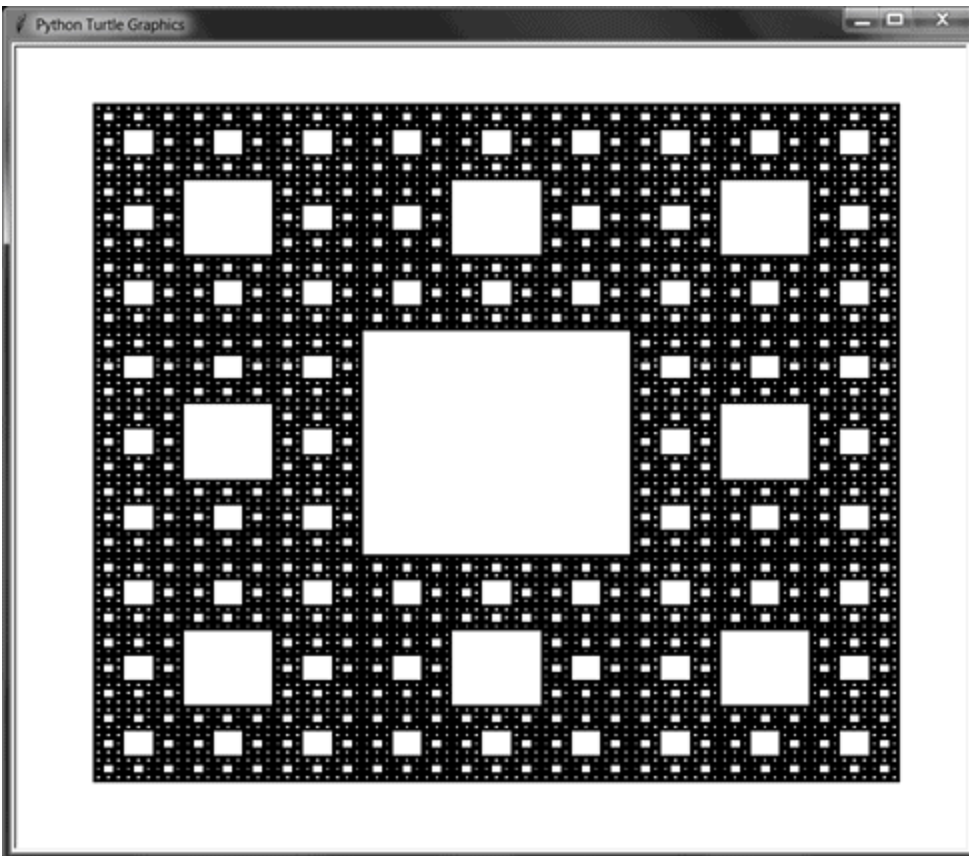


The Sierpiński Carpet

A fractal shape similar to the Sierpiński triangle can be drawn using rectangles instead. This pattern is known as the *Sierpiński carpet*. Imagine splitting a black rectangle into a 3×3 grid, then “cutting out” the center rectangle. Repeat this pattern in the surrounding eight rectangles of the grid. When this is done recursively, you end up with a pattern like [Figure 9-7](#).



[Figure 9-7](#): The Sierpiński carpet

The Python program that draws the carpet uses the `turtle.begin_fill()` and `turtle.end_fill()` functions to create solid, filled-in shapes. The lines that the turtle draws between these calls are used to draw the shape, as in [Figure 9-8](#).